

GRA-AGENT: Toward Boundary Subjectivity in AI via Conflict-Driven Regulation and Dynamic Multi-Objective Optimization

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Abstract

Contemporary AI systems typically optimize a single, pre-defined loss or reward function. While effective for narrow tasks, this approach faces fundamental limitations in long-term, autonomous interaction with complex environments. Real-world agents must simultaneously account for safety, energy consumption, accuracy, cost, social constraints, and uncertainty. A scalar reward obfuscates inherent conflicts between objectives, leading to brittle or unsafe behavior. This paper proposes an architectural framework—GRA-AGENT—that treats these conflicts not as errors to be masked but as measurable phenomena to be managed. The architecture integrates hormone-like global regulation, dynamic goal weighting, and a Generalized Reconciliation of Aims (GRA) functional. We formalize the agent’s state, introduce the GRA functional for conflict resolution, and propose methods for dynamic priority adjustment. The concept of *boundary subjectivity* is operationally defined. We outline verification protocols and discuss applications in drug discovery, robotics, EUV lithography, generative art, and autonomous manufacturing. Critical limitations and falsifiability conditions are explicitly addressed, emphasizing that stability and behavioral self-correction do not imply consciousness or guaranteed universal utility.

1 Introduction

Modern AI systems excel at optimizing a predetermined scalar objective. However, this paradigm is ill-suited for agents required to operate autonomously over extended periods. Real-world deployment necessitates the simultaneous management of safety, resource constraints, accuracy, cost, social norms, and environmental uncertainty. A single reward function inherently conflates these potentially conflicting goals. For instance, increasing a robot’s speed may elevate collision risk; boosting an industrial process’s power can increase defect rates; and optimizing a generative model for novelty can degrade coherence.

This paper argues that such conflicts should not be masked but measured and revisited. We propose an architecture that resonates with self-correcting agents, reflexive control, and multi-objective optimization. The core idea is that zero output error does not imply the absence of internal dynamics; hidden processes can persist and impact long-term stability. Our GRA-AGENT framework operationalizes this principle.

2 Mathematical Model

2.1 Agent State

Let the agent's state at time t be defined by the vector:

$$S_t = (x_t, m_t, h_t, g_t, u_t), \quad (1)$$

where:

- x_t : physical and sensory state,
- m_t : memory,
- h_t : hormone-like regulatory variables,
- g_t : current goals,
- u_t : uncertainty estimate.

The hormone-like vector captures slow, global regulatory influences:

$$h_t = (h_{\text{energy}}, h_{\text{damage}}, h_{\text{stress}}, h_{\text{novelty}}, h_{\text{social}}, h_{\text{recovery}}). \quad (2)$$

These are not biological hormones but computational regulators that affect multiple behavioral modules.

Their dynamics are given by:

$$\dot{h}_i(t) = -k_i(h_i(t) - h_i^*) + \alpha_i r_i(t) + \beta_i \xi_i(t), \quad (3)$$

where h_i^* is a baseline level, $k_i > 0$ is a mean-reversion coefficient, $r_i(t)$ is an environmental signal, $\xi_i(t)$ is noise, and α_i, β_i are sensitivity parameters.

2.2 GRA Functional

Let $c_i(S)$ measure a specific contradiction: a constraint violation, a goal misalignment, an excessive risk, or an inconsistency between memory and current perception. The GRA functional is defined as:

$$J(S) = \sum_i w_i c_i(S)^2 + \sum_{i,j} \lambda_{ij} d_{ij}(S)^2 + \mu R(S), \quad (4)$$

where $R(S)$ penalizes risk, safety violations, or excessive uncertainty, and d_{ij} represent cross-term conflicts.

The GRA update step is:

$$S_{t+1} = \arg \min_{S' \in C(S_t)} J(S'), \quad (5)$$

where $C(S_t)$ is the set of admissible structural changes. This set may include:

- altering goal priorities,
- abandoning a contradictory assumption,

- requesting additional observations,
- deferring to an external verifier,
- stopping an action,
- transferring decision-making to a human,
- changing the planning architecture.

2.3 Dynamic Goal Weights

The agent maintains N objectives $f_1, \dots, f_N$ with weights satisfying:

$$\omega_i(t) \geq 0, \quad \sum_{i=1}^N \omega_i(t) = 1. \quad (6)$$

The overall objective function is:

$$F(S, t) = \sum_{i=1}^N \omega_i(t) f_i(S) + \rho J(S). \quad (7)$$

The weight dynamics are governed by:

$$\dot{\omega}_i = \eta \left[q_i(S, t) - \sum_j \omega_j q_j(S, t) \right] - \gamma(\omega_i - \omega_i^*), \quad (8)$$

where q_i is the current criticality of goal i , η is the adaptation rate, and γ is a stabilizing force.

3 Boundary Subjectivity

We propose an operational definition of *boundary subjectivity*, not equating to proven consciousness. An agent exhibits boundary subjectivity if it can:

- autonomously choose between admissible goals,
- maintain a model of its own state,
- revise plans,
- explain changes in priorities while respecting safety constraints.

This definition is behavioral and structural, not phenomenological.

4 Verification

4.1 General Protocol

Verification should encompass four levels:

1. Verification: correctness of algorithm implementation.
2. Validation: model fidelity to the real problem.
3. Uncertainty Quantification: reliability of results.
4. Robustness Testing: behavior under perturbations.

Every experiment must include:

- a baseline model without GRA,
- a model with standard gradient control,
- a model with GRA and fixed weights,
- a model with GRA and dynamic weights,
- identical initial conditions,
- pre-defined metrics,
- independent repetitions,
- an open experiment log.

4.2 Stability Testing

After finding a state S^* , a perturbation is introduced:

$$S_0 = S^* + \epsilon v, \quad \|v\| = 1. \quad (9)$$

The transient is measured:

$$\Delta J(t) = J(S_t) - J(S^*). \quad (10)$$

Local stability requires:

$$\lim_{t \rightarrow \infty} J(S_t) \leq J(S^*) + \epsilon. \quad (11)$$

Additional checks include recovery time, maximum deviation, and sensitivity to noise.

4.3 Metrics

Proposed metrics:

$$M_1 = \frac{1}{T} \int_0^T J(S_t) dt, \quad (12)$$

$$M_2 = \max_{t \in [0, T]} J(S_t), \quad (13)$$

$$M_3 = \tau_{\text{recovery}}, \quad (14)$$

$$M_4 = P(\text{unsafe action}), \quad (15)$$

$$M_5 = H(\Omega), \quad (16)$$

$$M_6 = C_{\text{compute}}. \quad (17)$$

Improvement is only proven when quality, stability, and cost improve simultaneously.

5 Applications

5.1 Drug Discovery

For multi-objective drug design:

$$f(x) = (-f_1(x), f_2(x), -f_3(x), f_4(x)). \quad (18)$$

GRA-AGENT seeks a stable compromise region, requiring independent toxicity, pharmacokinetics, and reproducibility filters.

5.2 Scientific Modeling

In physics and biology, c_i may represent violations of conservation laws, discrepancies with experiments, or numerical instability:

$$J_{\text{science}} = w_1 E_{\text{data}}^2 + w_2 E_{\text{law}}^2 + w_3 E_{\text{uncertainty}}^2 + w_4 E_{\text{complexity}}^2. \quad (19)$$

The complexity term prevents overfitting.

5.3 Robotics

Internal signals (battery level, temperature, damage, etc.) are integrated. Simultaneous increases in h_{damage} and h_{stress} trigger a speed reduction and diagnostic mode.

5.4 EUV Lithography

A GRA agent can manage pattern error, defectivity, exposure time, and optic contamination, but cannot modify critical constraints without external confirmation.

5.5 Autonomous Manufacturing

The functional may be:

$$J = w_q Q^2 + w_s S^2 + w_e E^2 + w_r R^2 + w_c C^2, \quad (20)$$

where safety remains a hard constraint: $S(S_t) \leq S_{\text{max}}$.

5.6 Generative Art

In art generation:

$$J_{\text{art}} = w_1 C_{\text{coherence}} + w_2 C_{\text{novelty}} + w_3 C_{\text{theme}} + w_4 C_{\text{ethics}}. \quad (21)$$

Art often thrives on tension; the agent seeks a permissible region:

$$J_{\min} \leq J \leq J_{\max}. \quad (22)$$

6 Limitations and Falsifiability

Several limitations are acknowledged:

- The concept of “contradiction” must be operationalized to avoid being unfalsifiable.
- $J = 0$ is often unattainable; a practical interval criterion $J \leq \varepsilon$ is recommended.
- Minimizing J does not guarantee utility; an agent could freeze to achieve low J .
- Boundary subjectivity does not imply consciousness.
- Claims of guaranteed stability require rigorous proof; for general libraries, the term “verifiable stability” is more appropriate.

7 Reproducible Experiment Plan

Essential protocol elements:

1. Publish source code and configurations.
2. Fix random seeds and dependency versions.
3. Detail all parameters.
4. Compare at least four architectures.
5. Perform at least 30 independent runs.
6. Add noise, delays, and incomplete data.
7. Conduct ablation studies.
8. Report means, medians, variances, and confidence intervals.
9. Separate training, validation, and test scenarios.
10. Publish negative results.
11. Conduct independent safety audits.

8 Conclusion

We have presented the GRA-AGENT architecture, combining hormone-like regulation, dynamic multi-objective optimization, and a conflict-measuring functional. The core principle is that autonomous agents must maintain internal stability and re-prioritize dynamically, not merely optimize a scalar reward. While applicable to diverse domains, claims regarding consciousness, guaranteed safety, or true subjectivity require separate, rigorous proof. The proposed framework is a step towards more robust, transparent, and adaptable AI systems.

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